

Discovery Executive Summary

Project: test-frontend · Generated: 18/08/2026, 16:29:06

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The react-calculator repository is a minimal frontend-only Single Page Application built with React 18, Vite, and SCSS. The entire codebase consists of 2 JSX source files (88 total lines), 2 style files (100 total lines), and standard Vite configuration. There is **no backend layer** — no server-side code, no API endpoints, no database access, and no server-side routing. Because the standard architecture & design hotspots (H1–H9) target backend patterns (controllers, service layers, repositories, domain boundaries), they do not apply to this frontend-only repository. No additional architecture hotspots were observed: the codebase is small enough that its single-component structure is appropriate for the application's scope. Frontend-specific modernization concerns (component decomposition, state management patterns, etc.) are covered by the Frontend Modernization report (03-frontend-modernization).

1.1 Benchmark Ratings Summary

No backend layer detected — backend hotspots H1–H9 not applicable.

The repository contains only frontend code (React 18 SPA). The standard architecture & design hotspots (Fat Controllers, Missing Service Layer, Missing Repository Pattern, Circular Dependencies, Shared Utility Abuse, Direct SQL in Controllers, God Classes, Domain Boundary Violations, Shared Database Coupling) all target backend/server-side patterns and do not apply.

No additional hotspots beyond the standard set were observed.

The codebase is a 2-file, 88-LOC React application with a single component (`App.jsx` , 78 lines) and a standard entry point (`main.jsx` , 10 lines). At this scale, the absence of component decomposition, a service layer, or state management abstraction is architecturally appropriate. Frontend architecture concerns are covered by the Frontend Modernization report (03-frontend-modernization).

1.4 Actions Required

No architecture & design hotspots require action. All standard hotspots (H1–H9) are not applicable to this frontend-only repository, and no additional hotspots were observed.

1.5 Expected Outcomes

- Current state is architecturally sound for the application's minimal scope — a single React component with local state is the correct pattern for a small calculator widget.
- No refactoring needed today — the codebase is small enough that additional abstraction layers would add complexity without benefit.
- Growth-ready path is clear — if features expand (history, themes, scientific mode), the target architecture (component decomposition + service extraction + context-based state) is straightforward to adopt incrementally.
- No backend coupling risks — the application has no server-side dependencies, no API integrations, and no database access.
- Frontend-specific concerns (component size thresholds, state management patterns, style architecture) are tracked in the Frontend Modernization report (03-frontend-modernization).
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